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Global–local consistent semi-supervised segmentation of histopathological image with different perturbations ☆

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ABSTRACT

A histopathological image is a microscopic image applied to examine cellular and tissue structures and identify any abnormalities or disease processes. Histopathological image segmentation is a prerequisite step for analyzing histopathological images that can divide an image into meaningful regions or objects to accurately classify and analyze tissue structures, cellular regions, or particular histological entities. However, the existing deep learning based pathological image segmentation methods require huge annotation efforts from the pathologists, which is labor-intensive and time-consuming. In this scenario, it has become a hotspot to leverage abundantly available unlabeled data to help learn segmentation models given limited labeled data. In this paper, we propose a global–local consistent semi-supervised segmentation (GLCS) model that enforce the consistency of the segmentation results with weak and strong perturbations on unlabeled data. In GLCS, we firstly generate different weak perturbations for each unlabeled sample, and then add a regularization term to ensure the segmentation consistency among different weak perturbations. Next, different from the existing studies applying the regression methods to match the segmentation results among different perturbations, our methods are based on the generative adversarial learning that can keep the global structure consistency among unlabeled data with different strength of perturbations. Finally, we also add a patch-correlation based regularization term to preserve the local structure similarity among different perturbations images. We validate our GLCS on three datasets, i.e. Glas, Crag and MoNuSeg. The experimental results show that our method can achieve to the dice ratio of 90.35, 82.61 and 81.60 with 1:1 proportion of labeled data, which are significantly superior to the state-of-the-art semi-supervised histopathological image segmentation methods. Our code is public available at https://github.com/ISBELLAG/GLCS.

1. Introduction

It is widely recognized that histopathological images are generally considered as the golden standard for determining the aggressiveness of human cancers, since they can provide cell information that is highly related to the degree of the aggressiveness of cancers. Histopathological images are microscopic images of tissue samples obtained from biopsies or surgical specimens. These images are typically stained to enhance certain cellular or tissue structures, allowing the pathologists to evaluate and diagnose various diseases [1,2]. Given the digital pathological images, the segmentation of tissues and cells emerges as a crucial step in pathological image analysis. However, it is still a challenge to segment regions of interest (ROIs) from the histopathological images. For instance, for tissue segmentation, the edges of the tissues display a variety of shapes, and the gaps between adjacent tissues are also very narrow, making the segmentation between tissues extremely challenging (shown in Fig. 1A). As to the task of cell segmentation, the small size of cell targets and the occurrence of cell overlap (shown in Fig. 1B) further increase its complexity. Although manual segmentation is feasible, it is labor-intensive and time-consuming. Hence, how to make use of the machine learning algorithms to help reduce the workload of manual segmentation has become a new focus on computational pathology.

Recently, since deep learning has been extremely successful in the filed of computer vision on various nature image segmenting tasks [3–5], many researchers set out to apply deep learning algorithms to solve the pathological image segmentation problems. For instance, Wang et al. [6] designed a segmentation network based on channel

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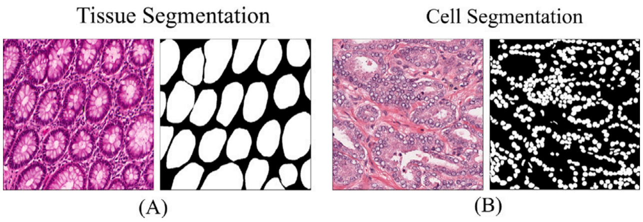

Fig. 1. Challenges for histopathological image segmentation. (A) Sample visualization for tissue segmentation, the edges of the tissues display a variety of shapes, and the gaps between adjacent tissue structures are very narrow. (B) Sample visualization for cell segmentation, the cells are with small size and the occurrence of cell overlap can also be observed.

attention module that can not only bridge the semantic and resolution gap in pathological images, but also captures more complex channel dependencies. Other studies include [7] have designed a two-stage learning framework by stacking two UNets network structure to segment nuclei from the histopathological images. All the results suggested that applying the deep learning technique can improve the accuracy on pathological image segmentation task. However, most of the above deep segmentation methods required huge amounts of labeling efforts from the pathologists, thus leading to high cost of both time and money.

To alleviate such problem, the semi-supervised semantic segmentation algorithms have been proposed [8]. In such learning paradigm, only a few images with annotations are provided, which aims at taking advantage of a large amount of unlabeled data. Generally, current semi-supervised segmentation algorithms can be categorized into two folds, *i.e.,* pseudo-label and consistent based methods. On one hand, the pseudo-label based methods assign the pseudo-label to the pixels in order to boost the performance on supervised segmentation model [9,10]. However, such pseudo-label based methods ignore part of unlabeled data during the training process that will affect the generalization ability of the segmentation model [11]. For solving this problem, the consistency learning based semi-supervised segmentation algorithms are proposed to enforces the consistency between the outputs from different views of unlabeled data. For example, Yassine et al. [12] apply different feature perturbation strategies to enforce the agreement between their semantic segmentation results and the output from the non-perturbed inputs. French et al. [13] explore model ensemble with image perturbations that can provide better training convergence and accuracy.

Even though much progress has been achieved on the consistency based semi-supervised segmentation model, most of them apply the regression algorithm to match the segmentation results among different perturbations with the same strength [13], which has limited capability in constructing robust segmentation models [14]. Inspired by the FixMatch model [15], which introduces the strong and weak perturbations on the unlabeled data to enforce the pixel-level consistency, we also increase the robustness of our model by incorporating perturbations of different strength. However, the aforementioned methods pay more attention on the pixel-level consistency rather than the global spatial contiguity within the segmentation results of different perturbations. In fact, similar to the natural images, we could also observe the global consistency on the histopathological images in terms of color and morphology. For instance, the lymphocytes usually have consistent morphology features that are characterized by the round shape with small size [16]. Also, the regression loss applied in the previous studies [17,18] have difficulty in exploring the local structure similarity among different perturbations that is crucial for improving the final segmentation performance. Based on the above considerations, we propose a Global–Local Consistent Semi-supervised segmentation (GLCS) model on the unlabeled data. Specifically, GLCS firstly considers the agreement of the segmentation results on different weak perturbations of the unlabeled samples. In addition, different from the existing adversarial learning studies that enforce the global consistency between the segmentation results and the groundtruth masks [19–21], we formulate the objective function of GLCS based on generative adversarial learning framework [22], which aims at producing the segmentation results from the strong perturbation that cannot be distinguished by the output of the weak perturbations for generating more robust segmentation results. Last but not the least, we further add the patch-correlation based regularization term on the traditional image-based adversarial learning framework to preserve the local structure similarity among different perturbation images.

Overall, the main contributions of this paper are presented as follows: (1) We propose a Global–Local Consistent Semi-Supervised Segmentation (GLCS) model that enforce the consistency of the segmentation results with weak and strong perturbations on the unlabeled data. (2) We apply the generative adversarial network to keep the global structure consistency among unlabeled images with different strength of perturbations. (3) We propose a novel regularization term based on patch correlation to maintain the local structural similarity between weak and strong perturbations for unlabeled data. (4) We evaluate our method on three datasets for different pathological image segmentation tasks. The experimental results indicate that our GLCS model can achieve superior segmentation performance than the comparing methods. More specifically, for the experimental results with 1:1 proportion of labeled data, the Dice score will be significantly improved by 4% and 2% for tissue and nuclei segmentation tasks, respectively.

The remainder of the paper is organized as follows, Section 2 introduces the related works; In Section 3, we illustrate our proposed GLCS algorithm. Then, we present the experimental results in Section 4. In Section 5, we discuss the effects of the parameters in our proposed GLCS. Finally, we summarize our work and point out the limitations of this study in Section 6.

2. Related work

In this section, we first summarize the SOTA deep learning algorithms for the segmentation of histopathological images. Then, we review the existing semi-supervised semantic segmentation algorithms that are related to our work.

2.1. Deep learning for histopathological image segmentation

In recent years, deep learning-based segmentation algorithms has been designed for histopathological image segmentation since the deep learning technology can better handle complex features in images, resulting in more accurate segmentation results. For instance, Chen et al. [23] propose an efficient deep contour-aware network that utilizes multi-scale contextual features and auxiliary supervision information for glandular tissue segmentation. In order to capture the multi-scale information of the to-be-segmented images, Wazir et al. [24] develop a simple but effective attention module on both encoder and decoder branches of the deep network to fuse the global and local features via residual links. Gorkem et al. [25] propose the idea of dual-cross attention to enhance the skip connections in the UNet architecture that can sequentially capture the channel and spatial dependencies across multiple scale encoder features. Moreover, Simon et al. [26] propose a fully convolutional neural network with spatial pyramid to perform effective multi-scale aggregation for histopathological image segmentation. However, most of the supervised segmentation methods require a large amount of precisely annotated images, which is labor-intensive and require huge annotation efforts from the experts.

2.2. Semi-supervised semantic segmentation

Semi-supervised learning is a powerful machine learning paradigm where a small portion of labeled data is used in conjunction with a large amount of unlabeled data to train the model. The main challenge in this approach is how to effectively utilize the rich unlabeled data. Currently, the existing semi-supervised semantic segmentation methods are mainly consisted of two categories *i.e.,* pseudo-labeling and consistency-based methods. The pseudo-labeling methods first train a model using the labeled data and then use this initial model to generate pseudo-labels for the unlabeled data, followed by retraining the model with both labeled and pseudo-labeled data to refine the model [27,28]. However, the pseudo-labeling methods are sensitive to noise, leading to unreliable segmentation results [29,30]. In contrast, the consistency-based methods [31] can improve model stability and robustness by enforcing the model to maintain consistency under different perturbations. For instance, Chen et al. [32] propose a new consistency based regularization term encouraging high similarity between the predictions of two different perturbed networks for the same input image. Luo et al. [33] propose a dual-task deep network that jointly learns pixel-level segmentation and task-level consistency regularization for object segmentation. Subsequently, they also consider the consistency among different perturbations of unlabeled data from a multiscale perspective [34]. Moreover, based on the Mean Teacher model, the studies in [17,18] add weak perturbations on the unlabeled data in order to achieve consistent outputs. However, in comparison with our GLCS, they only incorporate weak perturbations on the unlabeled data that has limited ability in building robust segmentation models. To address the above problem, The FixMatch model proposed in [15] introduce strong and weak perturbations to unlabeled data, using consistency regularization to constrain pixel-level segmentation results. However, in comparison with GLCS, FixMatch only adopts the pixel-level constrains that overlook the global consistency of boundary or tissues structure in the histopathological image that are particular important for deriving satisfied segmentation results. Recently, the adversarial learning strategy is also introduced to conduct the semi-supervised segmentation task [19–21] that are related to our study. Nevertheless, in comparison with our study, they only ensure the consistency between the segmentation results of labeled data and the ground truth that are unable to utilize the unlabeled data for model training. In addition, all these adversarial learning based segmentation models neglect to take the local structure information into consideration, and our GLCS incorporate a patch-correlation based regularization term to ensure that the local structure consistency among different types of perturbations could be captured.

3. Method

The overall architecture of the Global–local consistent semi-supervised segmentation (GLCS) proposed in this work is illustrated in Fig. 2. For the labeled data, we define a supervised segmentation module based on the UNet network. As to the unlabeled data, we introduce two types of perturbations with different strength (i.e., weak perturbation and strong perturbation). For the weak perturbation, we randomly add the Gaussian noise with the variance tuned from [0,0.1) on the unlabeled image, as to strong perturbations, the variance of Gaussian noise is randomly tuned from [0.1, 1]. Then, a weak perturbation consistent module is developed to enforce the segmentation consistency among different weak perturbations on the unlabeled data. Moreover, the weak-strong perturbation consistent module is designed to ensure the segmentation consistency between weak and strong perturbations. In weak-strong perturbation consistent module, the generative adversarial network is applied to keep the global structure consistency among unlabeled data, while a patch-correlation based regularization term is adopted to keep their local structure consistency. In the training process of GLCS, an alternative optimization strategy is adopted. Specifically, given the segmentation model G, the discriminator D is updated in order to distinguish segmentation results between the weak and strong perturbations. After the parameters in D is determined, we apply the gradient descent algorithm to optimize the parameters in G. In what follows, we will introduce them in detail:

3.1. Supervised segmentation module

Semi-supervised semantic segmentation aims to generalize from a combination set of pixel-wise labeled images $D^L = \{(x_i, y_i)\}_{i=1}^M$ and the unlabeled images $D^U = \{u_i\}_{i=1}^N$. In the segmentation network, we employ UNet as the backbone network, consisting of four encoders and four decoders. Each encoder consists of two 3×3 convolutional layers with ReLU activation followed by a 2×2 max pooling operation for downsampling. Each decoder is formed by an upsampling convolutional layer concatenated with two 3×3 convolutions layers and ReLU activation function, aiming to restore the high-level semantic features to the resolution of the input image. Skip connections are also employed between the encoder and decoder. Based on the UNet, We define the supervised segmentation loss of GLCS on the labeled data *i.e.,* L_s as follows:

$$L_s = L_{BCE} + L_{Dice} \tag{1}$$

$$L_{BCE} = -\frac{1}{M}\sum_{i=1}^{M}[y_i \log\big(G(x_i)\big) + (1-y_i)\log\big(1-G(x_i)\big)] \tag{2}$$

$$L_{Dice} = 1 - \frac{2\sum_{i=1}^{M} G(x_i)y_i}{\sum_{i=1}^{M} G(x_i)^2 + \sum_{i=1}^{M} y_i^2 + \varepsilon} \tag{3}$$

As shown from Eq. (1), L_s is formulated by the combination of L_{BCE} and L_{Dice}. Here, L_{Dice} and L_{BCE} referred to the Dice loss and BCE loss applied to measure the pixel-level inconsistency between ground truth y_i and the prediction values $G(x_i)$, and ε in L_{Dice} refers to the smoothing factor with the fixed value of 1e−5.

3.2. Weak perturbation consistent module

In the weak perturbation consistent module, we consider different weak perturbations for each unlabeled sample and define a consistency regularization term to ensure that the segmentation results among them are matched. Mathematically, suppose that u_i^w and $\overline{u_i^w}$ correspond to two weak perturbations of the unlabeled sample $u_i (i = 1, 2, \ldots, N)$. Then, the objective function for the weak perturbation consistent module can be defined as:

$$L_{wc} = \frac{1}{N}\sum_{i=1}^{N}(\left\|G(u_i^w) - G(\overline{u_i^w})\right\|_F^2)^2 \tag{4}$$

As can be seen from Eq. (4), for different weak perturbations on the unlabeled sample u_i, the consistency regularization enforce the segmentation model generate similar outputs, which is verified to improve the generalization ability of the segmentation model.

3.3. Weak-strong perturbation consistent module

In the weak perturbation consistent module, we only enforce the consistency among different weak perturbations. Recent studies also verify that producing perturbations with different strength can further improve the robustness of the semi-supervised learning model [15]. In this section, we leverage both weak and strong perturbations, where the weak perturbation aims to add small variations on the unlabeled samples while the strong perturbation will affect the characteristics or make major changes for the input images. We show the unlabeled images with strong and weak perturbations in Fig. 3. Different from the existing studies applying the regression algorithms to match the output of different types of perturbations that will miss the global structure consistency among them, we apply the generative adversarial learning

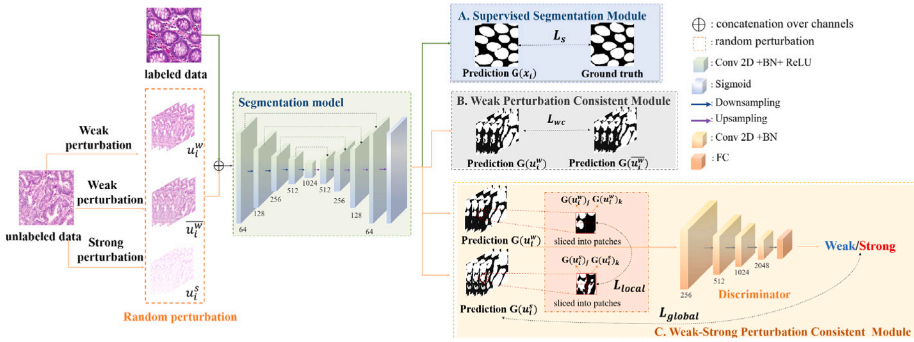

Fig. 2. The framework of the proposed GLCS method.

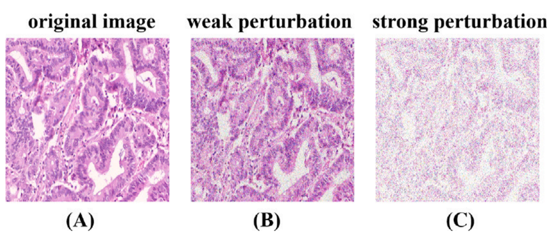

Fig. 3. The perturbation results for the unlabeled images. (A) The original unlabeled image. (B) The unlabeled image with weak perturbation. (C) The unlabeled image with strong perturbation.

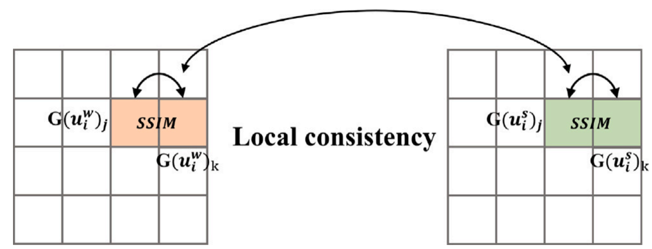

Fig. 4. Visualization for the calculation of patch-correlation based regularization term. $G(u_i^w)$ and $G(u_i^s)$ are the segmentation results for the weak and strong perturbations of the unlabeled sample u_i. $G(u_i^w)_j$ and $G(u_i^s)_j$ refer to the spitted jth patch for $G(u_i^w)$ and $G(u_i^s)$, respectively. We calculate the index of structural similarity index measure (SSIM) between the jth and the kth patches on both weak and strong perturbations, and hope their SSIM values are similar in order to keep the local consistency between weak and strong perturbations at local scale.

framework to ensure that the segmentation results from the strong perturbation cannot be distinguished by the segmentation output of the weak perturbations. In other words, rather than assess the consistency of perturbations at pixel-level, we consider the global spatial contiguity of different types of perturbations. Specifically, given $G(u_i^w)$, $G(u_i^S)$ as the segmentation results for the weak and strong perturbations of the unlabeled sample u_i, respectively. The objective function characterizing the global consistency *i.e.,* L_{global} between them can be formulated as:

$$\min_G \max_D L_{global}(G, D) = E_{u_i^w \in Q^w, u_i^s \in Q^s} \log D(G(u_i^w)) + \log(1 - D(G(u_i^s))) \tag{5}$$

where Q_w and Q_s represent the sets of samples with weak and strong perturbations, respectively. Here, the discriminator D is applied to distinguish the segmentation results between weak and strong perturbations for each of the unlabeled samples. On the other hand, $G(u_i^S)$ aims at generating the segmentation results from the strong perturbation that cannot be distinguished by the output from the weak perturbation. Thus, the global spatial and contextual information among different strength of perturbations can be preserved, leading to more accurate and robust segmentation results for semi-supervised learning.

Furthermore, to address the problem of under-utilization of local information in generative adversarial networks, we devise a patch-correlation based regularization term that is dedicated to maintaining the intrinsic local similarity between the segmentation results of weak and strong perturbations. Specifically, as shown in Fig. 4, we sliced the outputs of the unlabeled sample u_i under different strength of perturbations i.e., $G(u_i^w)$ and $G(u_i^s)$ into N equal-sized patches. Then, the patch-correlation based regularization term can be defined as:

$$L_{local}(G(u_i^w), G(u_i^s)) = \sum_{j=1}^{N} \sum_{k=j+1}^{N} \left[f\left(G(u_i^w)_j, G(u_i^w)_k\right) - f(G(u_i^s)_j, G(u_i^s)_k) \right]^2 \tag{6}$$

where $G(u_i^s)_j$ and $G(u_i^w)_j$ refer to the splitted jth patch for $G(u_i^s)$ and $G(u_i^w)$, respectively. In addition, $f(G(u_i^s)_j, G(u_i^s)_k)$ is the similarity calculation function, and we apply the index of structural similarity index measure(SSIM) that can restore the structure similarity information between weak and strong perturbation at local scale. Finally, the objective function of the proposed GLCS model can be formulated by the combination of Eqs. (1), (4), (5) and (6), which is shown as follows:

$$\begin{aligned} \min_G \max_D L_{se}(G, D) = {} & \lambda_{global} E_{u_i^w \in Q^w, u_i^s \in Q^s} (\log D(G(u_i^w)) \\ & + \log(1 - D(G(u_i^s))) \\ & + \lambda_{local} L_{local} \left(G(u_i^w), G\left(u_i^s\right)\right) + \lambda_{wc} L_{wc}) + \lambda_s L_s \end{aligned} \tag{7}$$

where λ_{global}, λ_{local}, λ_{wc} and λ_s are regularization parameters need to be tuned via cross-validation.

3.4. Optimization

The overall training scheme of the proposed framework is described in Algorithm 1. For the optimization problem shown in Eq. (7), we can adopt the alternative strategy to optimize the parameters ω_G and ω_D in G and D, respectively. More specifically, given the segmentation model G, the optimization problem for D is formulated as:

$$\max_D E_{u_i^w \in Q^w, u_i^s \in Q^s} \log D(G(u_i^w)) + \log(1 - D(G(u_i^s))) \tag{8}$$

which can be optimized by the gradient ascent algorithm shown below:

$$g_{\omega_D} \leftarrow \nabla_{\omega_D} \frac{1}{N} \sum_{i=1}^{N} \left[log D(G(u_i^w)) + log(1 - D(G(u_i^s))) \right] \tag{9}$$

$$\omega_D \leftarrow \omega_D + \eta_d \mathbf{SGD}(\omega_D, g_{\omega_D}) \tag{10}$$

where η_d represents the learning rate for the discriminator D. After ω_D is determined, we apply the gradient descent algorithm shown in

Eqs. (11) and (12) to update the parameters in the generator *i.e.,* ω_G.

$$g_{\omega_G} \leftarrow \nabla_{\omega_G}(\frac{1}{N}\sum_{i=1}^{N}(\lambda_{global}(\log D(G(u_i^w)) + \log(1 - D(G(u_i^s)))) + \lambda_{local}L_{local} + \lambda_{wc}L_{wc}) + \lambda_s L_s) \tag{11}$$

$$\omega_G \leftarrow \omega_D - \eta_g \mathbf{ADAM}(\omega_G, g_{\omega_G}) \tag{12}$$

where η_g represents the learning rate for the generator G. The detailed optimization procedure of our GLCS model is shown below.

Algorithm 1 The optimization procedure for GLCS.

Input:
Labeled training set: $D^L = \{(x_i, y_i)\}_{i=1}^M$, unlabeled training set: $D^U = \{u_i\}_{i=1}^N$.
Output:
The parameters for the segmentation model i.e., ω_G. The parameters for the discriminator i.e., ω_D.

1: **while** (iter < Max iteration) **do**
2: Sampled n_u samples *i.e.,* D_s^U from D^U .
3: Sampled n_l samples *i.e.,* D_s^L from D^L.
4: Generate n_p weak perturbations for each sample in D_s^U.
5: Generate n_p strong perturbations for each sample in D_s^U.
6: Update ω_D according to Eq. (9) and Eq. (10).
7: Calculate the supervised segmentation loss L_s according to Eq. (1).
8: Calculate the local consistent loss L_{local} according to Eq. (6).
9: Calculate the weak perturbation consistent loss L_{wc} according to Eq. (4).
10: Update ω_G according to Eq. (11) and Eq. (12).
11: **end while**

4. Experiments and results

In this section, we firstly provide an overview of the datasets used in this study in Section 4.1, followed by the introduction of the experimental settings in Section 4.2. In Section 4.3, we introduce the evaluation metrics for the comparison of different methods. Then, we conduct the quantitative comparisons among different methods for tissue and cell segmentation in Sections 4.4 and 4.5, respectively. Next, we visualize the segmentation results among different methods in Section 4.6. Finally, we display the results for ablation studies in Section 4.7.

4.1. Dataset

In this study, we choose three datasets i.e., Glas, Crag, and MoNuSeg datasets in this study since they are most widely used to evaluate the performance for histopathological image segmentation task [6,35,36]. Specifically, the Glas dataset is consisted of 165 image patches [37] derived from 16 whole-slide histopathological images for the purpose of gland tissue segmentation. The size of each image patch are with the size of 775 × 522. Moreover, the Crag dataset contains 213 H&E colorectal adenocarcinoma image tiles at 20x magnification with full tissue-level annotation, and all images in Crag dataset are with the size of 1512 × 1516 [38]. Finally, the MoNuSeg dataset [25] contains thirty cell-level annotated patches with the size of 1000 × 1000. We have provided a detailed overview of the main characteristics of the three datasets (*i.e.,* Glas, Crag and MoNuseg) in Table 1.

4.2. Experimental settings

In our experiment, we follow the studies in [6,35,36], which randomly divided the whole dataset into the training set, validation set and testing set, and the sample size for each set is shown in Table 2. In addition, we used random data enhancement (*i.e.,* flip and rotation) on each sample to prevent over-fitting. For the images in the training set, we randomly split it into labeled and unlabeled parts with different proportions (*i.e.,* 1:1 and 1:3). Then, we generate the weak and strong perturbations of the unlabeled data by adding the Gaussian noise with different variance values. Specifically, for the weak perturbation, we randomly add the Gaussian noise with the variance tuned from [0,0.1) on the unlabeled image, as to strong perturbations, the variance of Gaussian noise is randomly tuned from [0.1, 1]. Besides, for each unlabeled image, we apply the weak or strong perturbation strategy to generate 20 images with perturbation, and use them train the proposed GLCS model. For the supervised segmentation task, the UNet [39] is adopted, and the ResNet50 is applied to distinguish the segmentation results from weak and strong perturbations. We show the parameter settings in our experiments in Table 3. Our experiments are conducted on a single RTX 3090. In terms of running time, the proposed GLCS takes 7.9 h, 11.8 h, and 4.3 h for model training on Glas, Crag, and MoNuSeg, respectively if the ratio between the number of labeled and unlabeled images is set as 1:1. In the training process of GLCS, the input image size was set to 224 × 224 with the batch size of 2. To ensure the comparability of our experimental results, we use the same image size and batch size for other comparing methods. Additionally, we apply an early stop mechanism where training procedure will stop when the parameters of the network remains unchanged for 50 epochs. The segmentation model uses the Adam optimizer with an initial learning rate of 0.00004, while the discriminator uses the SGD optimizer with an initial learning rate of 0.0004. Then, we utilize the cosine annealing warm restarts algorithm to update the learning rate [40]. Here, the number of iterations for the first restart is set to 10, and the length of restart cycles gradually increases by a factor of 1.

Table 1
The main characteristics of Glas, Crag and MoNuSeg.

Dataset	Image type	Size	Color system	Quantity	Origin
Glas	bmp	775 × 522	RGB	165	[37]
Crag	png	1512 × 1516	RGB	213	[38]
MoNuSeg	tif	1000 × 1000	RGB	44	[25]

Table 2
The sample size for the training, validation and testing sets across different datasets.

Dataset	Trainning set	Validation set	Testing set
Glas	99	33	33
Crag	127	42	44
MoNuSeg	30	7	7

Table 3
The main settings/parameters of the proposed method.

Parameters	Value
Image Size	224 × 224
Batch Size	2
Early Stopping	50 epochs
Learning rate for the segmentation model	0.00004
Learning rate for the discriminator	0.0004
Optimizer for the segmentation model	ADAM
Optimizer for the discriminator	SGD

4.3. Evaluation metrics

In our experiment, we use the Dice coefficient (Dice) and Intersection over Union (IoU) to measure the performance of all segmentation methods on the Glas, Crag, and MoNuSeg datasets. Specifically, the Dice coefficient is defined as follows:

$$Dice = \frac{2\,|G(x)| \cap |y|}{G(x) + y} \tag{13}$$

where $G(x)$ and y represent the segmentation result and the ground truth, respectively. Furthermore, we also use Intersection over Union (IoU) to evaluate our segmentation results, which is defined as follows:

$$IoU = \frac{G(x) \cap y}{G(x) \cup y} \tag{14}$$

4.4. Quantitative comparisons of different methods for tissue segmentation

In this study, we compare GLCS with the following seven semi-supervised segmentation methods with different proportions of labeled data by the measurement of Dice Ratio and IoU ratio: (1) **MT(2017)** [17]: A semi-supervised segmentation method composed of student model and teacher model, where the student model tries to mimic the teacher model's predictions on the unlabeled data. (2) **UA-MT (2019)** [18]: A uncertainty-aware semi-supervised segmentation scheme based on the Mean Teacher framework. (3) **S4GAN (2019)** [21]: A semi-supervised semantic segmentation framework utilizing the generative adversarial network to enhance the consistency between the segmentation results and the ground truth mask. (4) **FixMatch (2020)** [15]: A adversarial semi-supervised learning framework by the combination of the consistency and pseudo-labeling methods. (5) **SASS-Net(2020)** [19]: A shape-aware semi-supervised segmentation approach utilizing the geometric constraints and generative adversarial learning to improve the consistency of geometric representations between labeled and unlabeled images. (6) **DTC(2021)** [33]: A novel dual-task-consistency semi-supervised segmentation framework jointly predicts the pixel-wise segmentation map and the geometry-aware level set representation of the target. (7) **URPC (2022)** [34]: A semi-supervised segmentation model considering the consistency among the perturbations of unlabeled data.

Tables 4 and 5 present the tissue segmentation results of GLCS and the above comparing methods on Glas and Crag datasets, respectively. As shown in Tables 4 and 5, in comparison with the supervised segmentation model only relying on the limited labeled image (first row in Tables 4 and 5), the utilization of the unlabeled image to train the semi-supervised segmentation model can effectively improve the segmentation results. Moreover, in comparison with the upper and lower part of Tables 4 and 5, we observe that the incorporation of more labeled images can benefit the segmentation task for all semi-supervised segmentation methods. In addition, we find that our method i.e., GLCS can achieve consistently higher Dice ratio and IoU values than all comparing semi-supervised segmentation algorithms with different quantitative proportions between labeled and unlabeled images. Specifically, our GLCS is significantly superior to MT and UA-MT that are based on the teacher-student learning framework, showing the advantage of taking the consistency among different perturbations for semi-supervised learning. Secondly, we note that the proposed GLCS and other adversarial learning based segmentation models (i.e., S4GAN and SASS-Net) usually yield better segmentation performance than other consistency based semi-supervised learning algorithms. This is because the adversarial learning strategy could produce more realistic segmented images with global consistency than the comparing methods [21]. Finally, we notice that GLCS can achieve to the best segmentation performance among the three adversarial learning based semi-supervised learning methods (i.e., GLCS, S4GAN and SASS-Net). The reason lies in the fact that the proposed GLCS not only considers the global consistency among the perturbations of different strength but also take the local structure similarity among different perturbations images into consideration.

4.5. Quantitative comparisons of different methods for nuclei segmentation

To further evaluate the performance of our GLCS, we implement it on the MoNuSeg dataset aimed at segmenting nuclei from the histopathological images, and the results are shown in Table 6. As can be observed from Table 6, the proposed GLCS can achieve to the best dice ratio (i.e., 77.45) and the IoU value (i.e., 63.25) among all comparing methods if the proportion of labeled and unlabeled images is 3 to 1. As the proportion of the labeled images increased from 1:3 to 1:1, our method can still achieve to superior segmentation results (with a dice ratio of 81.60 and an IoU of 68.95). Similar to the reasons mentioned in Section 4.4, the better segmentation performance of our method can be attributed to the following two reasons. Firstly, we for the first time apply the adversarial learning strategy to enforce the consistency among different strength of perturbations that can effectively improve the performance and robustness of the semi-supervised learning models. Secondly, we also take the local consistency among different perturbations into consideration that can further benefit the segmentation task on histopathological images.

Table 4
Comparison of segmentation performance of different semi-supervised methods on Glas dataset. The upper part of the table presents experimental results with the 1:3 proportion of labeled data, while the lower part of the table presents experimental results with the 1:1 proportion of labeled data.

Method	Image Used		Metrics	
	Labeled	Unlabeled	Dice (%)↑	IoU (%)↑
UNet [39]	24(1:3)	0	76.87	64.46
MT [17]	24(1:3)	75	81.13	70.56
UA-MT [18]	24(1:3)	75	81.99	70.96
FixMatch [15]	24(1:3)	75	83.51	73.70
URPC [34]	24(1:3)	75	82.60	70.91
DTC [33]	24(1:3)	75	83.53	72.87
S4GAN [21]	24(1:3)	75	83.98	73.55
SASS-Net [19]	24(1:3)	75	84.05	74.17
GLCS(ours)	**24(1:3)**	**75**	**86.14**	**76.51**
UNet [39]	49(1:1)	0	83.19	73.50
MT [17]	49(1:1)	50	83.96	74.28
UA-MT [18]	49(1:1)	50	84.35	75.30
FixMatch [15]	49(1:1)	50	85.19	75.63
URPC [34]	49(1:1)	50	84.59	74.62
DTC [33]	49(1:1)	50	85.49	76.27
S4GAN [21]	49(1:1)	50	85.52	76.37
SASS-Net [19]	49(1:1)	50	86.01	76.72
GLCS(ours)	**49(1:1)**	**50**	**90.35**	**82.94**

Table 5
Comparison of segmentation performance of different semi-supervised methods on Crag dataset. The upper part of the table presents experimental results with the 1:3 proportion of labeled data, while the lower part of the table presents experimental results with the 1:1 proportion of labeled data.

Method	Image Used		Metrics	
	Labeled	Unlabeled	Dice (%)↑	IoU (%)↑
UNet [39]	33(1:3)	0	70.48	70.48
MT [17]	33(1:3)	94	73.98	60.87
UA-MT [18]	33(1:3)	94	72.95	60.10
FixMatch [15]	33(1:3)	94	74.81	63.56
URPC [34]	33(1:3)	94	74.44	62.30
DTC [33]	33(1:3)	94	74.59	61.93
S4GAN [21]	33(1:3)	94	73.96	61.87
SASS-Net [19]	33(1:3)	94	75.24	63.16
GLCS(ours)	**33(1:3)**	**94**	**80.29**	**69.42**
UNet [39]	63(1:1)	0	73.21	61.07
MT [17]	63(1:1)	64	74.36	63.12
UA-MT [18]	63(1:1)	64	76.07	64.51
FixMatch [15]	63(1:1)	64	77.21	65.98
URPC [34]	63(1:1)	64	76.34	65.03
DTC [33]	63(1:1)	64	77.54	66.39
S4GAN [21]	63(1:1)	64	76.06	64.85
SASS-Net [19]	63(1:1)	64	77.87	66.40
GLCS(ours)	**63(1:1)**	**64**	**82.61**	**72.37**

4.6. Qualitative comparisons of different methods for histopathological image segmentation

In order to further assess the segmentation performance of different methods, we also visualize the tissue and nuclei segmentation results

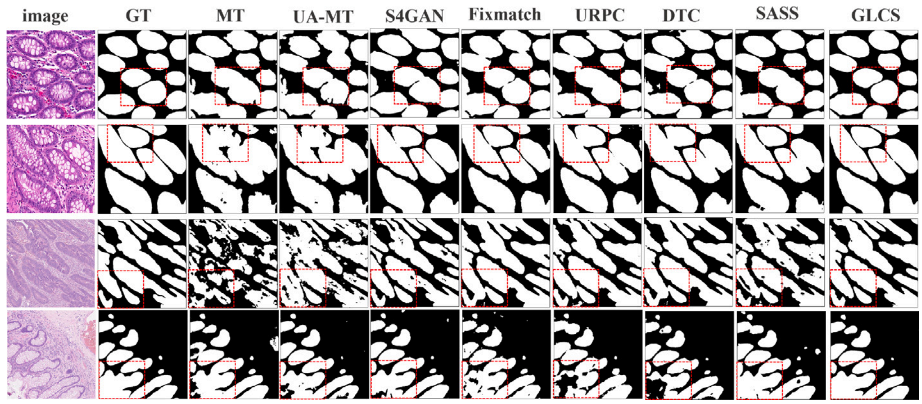

Fig. 5. Visualization results of several semi-supervised segmentation methods trained with 1:1 proportion of labeled data (The first two rows are from the Glas dataset, and the last two rows are from the Crag dataset.) As shown in the highlight part (with red frame) of the segmentation results, our method is more reliable in segmenting proximity tissues (shown in the first two rows). Moreover, our GLCS also shows its advantage in keeping the local structure of tissue for pathological image segmentation.(shown in the last two rows).

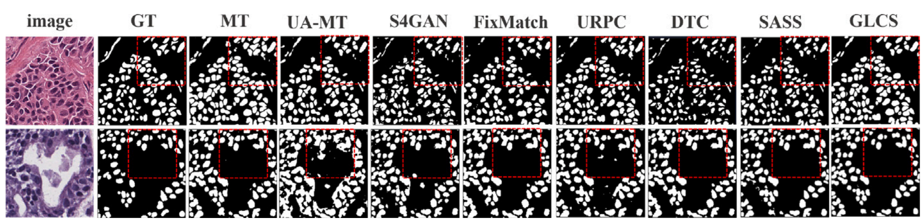

Fig. 6. Example results of several semi-supervised segmentation methods trained with 1:1 proportion of labeled data on the cell nucleus dataset (MoNuSeg dataset).

Table 6
Comparison of segmentation performance of different semi-supervised methods on MoNuSeg dataset. The upper part of the table presents experimental results with the 1:3 proportion of labeled data, while the lower part of the table presents experimental results with the 1:1 proportion of labeled data.

Method	Image used		Metrics	
	Labeled	Unlabeled	Dice (%)↑	IoU (%)↑
UNet [39]	8(1:3)	0	60.48	50.41
MT [17]	8(1:3)	22	67.01	50.49
UA-MT [18]	8(1:3)	22	70.63	55.47
FixMatch [15]	8(1:3)	22	70.87	55.13
URPC [34]	8(1:3)	22	72.38	60.44
DTC [33]	8(1:3)	22	72.53	57.52
S4GAN [21]	8(1:3)	22	72.45	57.07
SASS-Net [19]	8(1:3)	22	73.71	58.70
GLCS(ours)	**8(1:3)**	**22**	**77.45**	**63.25**
UNet [39]	15(1:1)	0	73.21	61.07
MT [17]	15(1:1)	15	76.83	63.28
UA-MT [18]	15(1:1)	15	76.32	60.48
FixMatch [15]	15(1:1)	15	77.24	62.85
URPC [34]	15(1:1)	15	77.09	62.72
DTC [33]	15(1:1)	15	78.80	65.20
S4GAN [21]	15(1:1)	15	77.26	63.07
SASS-Net [19]	15(1:1)	15	79.01	65.66
GLCS(ours)	**15(1:1)**	**15**	**81.60**	**68.95**

with a 1:1 proportion of labeled data in Fig. 5 and Fig. 6, respectively. As shown in Fig. 5, our method (GLCS) can achieve better segmentation performance than the comparing methods on Glas and Crag datasets for tissue segmentation especially when the individual components are touching together (the first two rows in Fig. 5). In addition, in comparison to other comparative methods (i.e., MT, UA-MT) that overlooks the global information of the image, our method is based on the adversarial learning framework for deriving global consistency segmentation results, which can more effectively identify the entire tissue structure. Moreover, as indicated by Fig. 6, our method can also better segment small components (i.e., small nuclei) with complex local structure than the comparing methods (i.e. SASS-Net, S4GAN), which further validate the advantage of using local patch-based structural regularization term for improving the segmentation results on pathological images.

4.7. Ablation study

To further evaluate the effectiveness of GLCS, we compare it with the their following variants based on the measurements of Dice ratio and IoU in Table 7. As shown in Fig. 7, (1) GLCS_V1 : A variant of GLCS, which only incorporate the weak perturbation consistent module (shown in Eq. (4)) for semi-supervised learning. (2) GLCS_V2 :A variant of GLCS, which contains both weak perturbation consistent module (shown in Eq. (4)) and Weak-Strong perturbation Consistent module, but neglect to take the patch-correlation based regularization term (shown in Eq. (6)) into consideration. (3) GLCS_V3 : A variant of GLCS, which applies the L2 norm to enforce the consistency between the segmentation results of weak and strong perturbations.

As can be observed from Table 7, both GLCS_V2 and GLCS outperform GLCS_V1 and GLCS_V3 under different proportions of labeled data. This is because we introduce the generative adversarial loss to ensure the consistency among different strength of perturbations from the global view, and thus lead to a more robust and accurate segmentation result. Moreover, GLCS can achieve better segmentation results than GLCS_V2. This is because GLCS takes into account the consistency of segmentation results from both weak perturbations and strong perturbations at local levels.

5. Discussion

In this section, to further demonstrate the effectiveness and rationality of our proposed GLCS model. Specifically, we first examine the impact of regularization parameters on different modules. Then, we investigate the influence of the number of perturbations applied on each unlabeled image.

Table 7
Ablation studies of the proposed method on Glas dataeset, Crag dataset and MoNuSeg dataset.

Method	Glas				Crag				MoNuSeg			
	1:3		1:1		1:3		1:1		1:3		1:1	
	Dice (%)	IoU (%)	Dice (%)	IoU (%)	Dice (%)	IoU (%)	Dice (%)	IoU (%)	Dice (%)	IoU (%)	Dice (%)	IoU (%)
GLCS_V1	81.27	70.45	83.27	74.37	74.95	61.92	76.97	64.87	70.29	55.21	76.73	63.11
GLCS_V2	84.06	75.01	86.49	76.20	78.12	66.77	80.26	70.87	73.62	58.54	78.26	64.91
GLCS_V3	83.12	71.10	84.95	75.82	75.38	62.82	79.02	67.39	71.76	56.31	77.87	64.01
GLCS	**86.14**	**76.51**	**90.35**	**82.94**	**80.29**	**69.42**	**82.61**	**72.37**	**77.45**	**63.25**	**81.60**	**68.95**

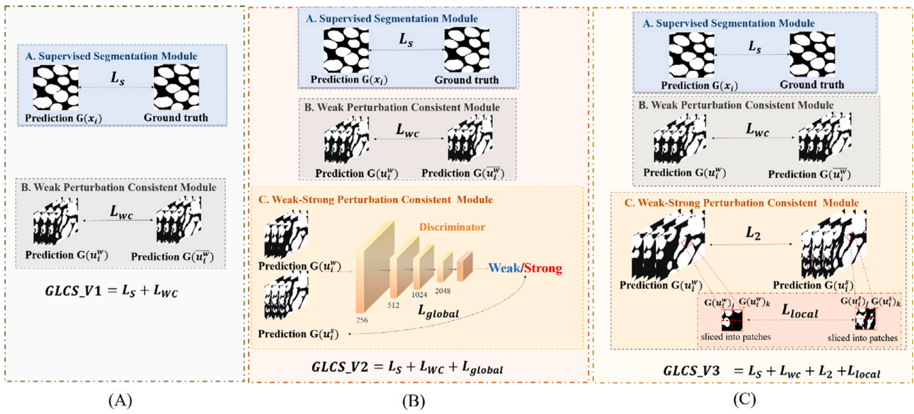

Fig. 7. Illustration of the variants of GLCS. (A) GLCS_V1: A variant of GLCS, which only incorporates the weak perturbation consistent module (shown in Eq. (4)) for semi-supervised learning. (B) GLCS_V2: A variant of GLCS, which contains both weak perturbation consistent module (shown in Eq. (4)) and weak-strong perturbation consistent Module, but neglect to take the patch-correlation based regularization term into consideration. (C) GLCS_V3: A variant of GLCS, which applies the L2 norm to enforce the consistency between the segmentation results of weak and strong perturbations.

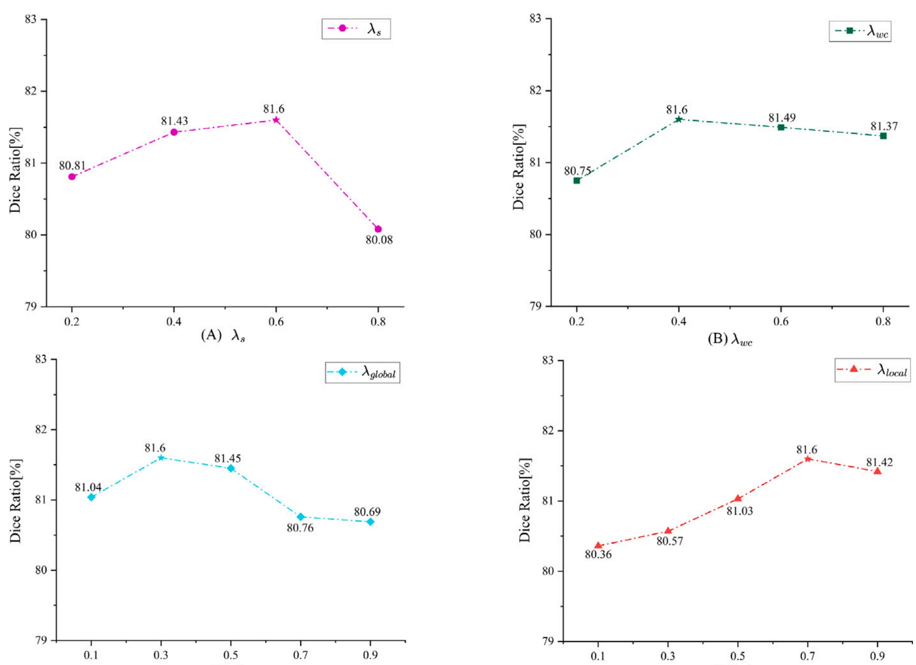

Fig. 8. The effects of different regularization parameters (i.e., $\lambda_s, \lambda_{wc}, \lambda_{global}, \lambda_{local}$) for all the modules in GLCS. (A) The effect of λ_s on GLCS. (B) The effect of λ_{wc} on GLCS. (C) The effect of λ_{global} on GLCS. (D) The effect of λ_{local} on GLCS.

5.1. Effects of different parameter settings for histopathological image segmentation

To investigate the impact of each module in GLCS, we discuss the effects of the regularization parameters (i.e., $\lambda_s, \lambda_{wc}, \lambda_{global}, \lambda_{local}$) for all the modules. Here, we discuss the impact of changing one parameter while keeping the remaining parameters unchanged. We show the experimental results on the MoNuSeg dataset with the labeled ratio of 1:1 in Fig. 8.

As shown in Fig. 8(B), the Dice values are generally stable when the parameters (*i.e.,*$\lambda_s, \lambda_{wc}, \lambda_{global}, \lambda_{local}$) are tuned in a certain range. These results indicate that our GLCS are robust to these parameters. In addition, although the dice ratio fluctuate in a small range with respect to the change of the regularization parameters, it is obvious that most of

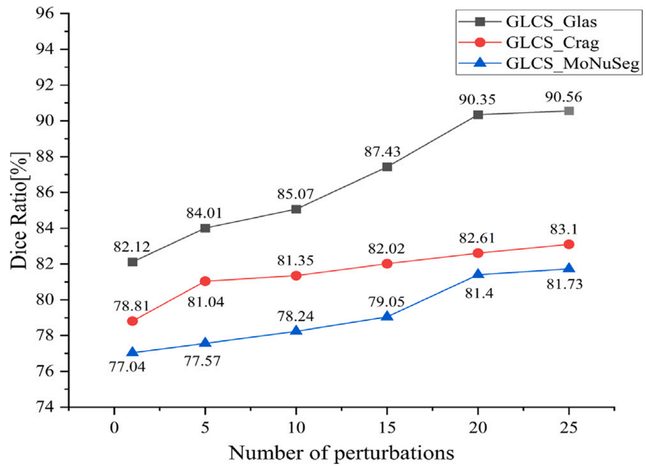

Fig. 9. The segmentation performance of the GLCS model on the Glas dataset, Crag dataset and MoNuSeg dataset with the proportion is 1:1 under different numbers of perturbations.

the inner intervals of the curves have relative large values of dice ratio than the two vertices, indicating that selecting a proper regularization parameter can help improve the segmentation performance of our GLCS model slightly.

5.2. Effects of the number of perturbations for each unlabeled image

In this section, we discuss the effects of the number of perturbations for each unlabeled data for Glas, Crag and the MoNuseg datasets, and the results are shown Fig. 9. As shown in Fig. 9, we observe that the segmentation performance will be improved as the number of perturbations for each unlabeled data increases, which shows the advantage of involving more perturbations for adversarial learning. On the other hand, we also observed that the segmentation results will be stable if the perturbation number for each sampled is larger than 20. These results indicate that the GLCS model is insensitive to the number of perturbations if enough perturbations are involved.

In this section, we discuss the effects of regularization parameters and the number of desired perturbations for each unlabeled image. In summary, our model is robust to all the regularization parameters (shown in Eq. (7)). One possible reason is that all of these four regularization terms are essential for generating desired segmentation results, and thus the occurrence of small variation on each regularization parameter will not largely affect the final results. On the other hand, the segmentation results of our model is stable if desirable number of perturbations are provided. This is because the performance of the adversarial learning is also influenced by the number of input samples, and our model will be robust to the number of perturbations given enough pairs of weak/strong perturbation samples.

6. Conclusion

In this paper, we propose a semi-supervised segmentation model based on global–local consistency, called GLCS. Experimental results on Glas, Crag and MoNuSeg datasets show that our GLCS method achieves the best segmentation performance than several state-of-the-art semi-supervised learning methods for different pathological image segmentation tasks. In summary, our model can effectively enhance the consistency of the segmentation results by the incorporations of strong and weak perturbations on the unlabeled data. By introducing the generative adversarial networks, the model focuses more on deriving global consistent segmentation results. Additionally, we add patch-based regularization term considering the similarity of local structures under different perturbations, thereby improving the performance of the segmentation model at local levels. In practical clinical research, our research can help pathologists analyze specific types of glandular structures and cell nuclei, enabling precise diagnosis and treatment of various types of cancers. Although much progress have been achieved, our GLCS also has three limitations. Firstly, the proposed GLCS gains the worst segmentation results on the MoNuSeg dataset since it is with the minimum sample size for the unlabeled images. This inspires us to design more perturbation or data augmentation ways that can increase the size and diversity of the unlabeled data for further performance improvement. Secondly, in this work, we use the basic UNet network to train the backbone segmentation model. Recently, many SOTA medical image segmentation models are proposed (i.e., UctransNet [6]). In future work, we will design more advanced segmentation model for histopathological image segmentation task. Thirdly, our experimental results indicate that the segmentation performance will be improved with the increased perturbations on each unlabeled data. However, the involvement of more perturbations will increase the burden for modal training. In future, it is desirable to design a filtering strategy to select the most valuable unlabeled samples for perturbations that can reduce the computation cost while maintain the satisfied segmentation results. Although our method can achieve satisfied segmentation results, the utilized pathological images for model training is only involved with small portion of cancer subtypes, it is still challenge the to generalize the segmentation model on the pan-cancer datasets for practical clinical application.

CRediT authorship contribution statement

Xi Guan: Methodology. **Qi Zhu:** Methodology. **Liang Sun:** Conceptualization. **Junyong Zhao:** Validation. **Daoqiang Zhang:** Writing – review & editing. **Peng Wan:** Visualization. **Wei Shao:** Writing – original draft, Supervision.

Declaration of competing interest

All the authors declare no interests with other people or organizations that could inappropriately influence this study

Data availability

Data will be made available on request.

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